

LouRIM Seminar Series on AI

AI Tools for Research

Seminar 1 of 4

*Enhancing scientific workflows
with artificial intelligence*

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UCLouvain



LOUVAIN
RESEARCH INSTITUTE
IN MANAGEMENT
AND ORGANIZATIONS

Publish-or-perish ?

« Scientists who engage in AI-augmented research
publish ? times more papers and
receive ? times more citations »

Publish-or-perish ?

« Scientists who engage in AI-augmented research
publish 3.02 times more papers and
receive 4.84 times more citations »

Hao, Qian Yue, et al. (2026) - *Nature*

One of the biggest drawback of AI in research
is not that it is **wrong...**

One of the biggest drawback of AI in research
is not that it is **wrong...**

...but that it is **convincingly wrong.**

La rectrice de l'UGent, Petra De Sutter, a utilisé des citations inventées par l'IA dans un discours



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03

Better Prompting

Context
+
Goal
+
Constraints
=
Better results

04

Sustainable AI



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Theoretical Presentation

ACT 1 — The Seduction

Let's publish a paper during this presentation

ACT 2 — Getting up to Speed

AI - ML - LLM - GenAI - RAGs - Agents

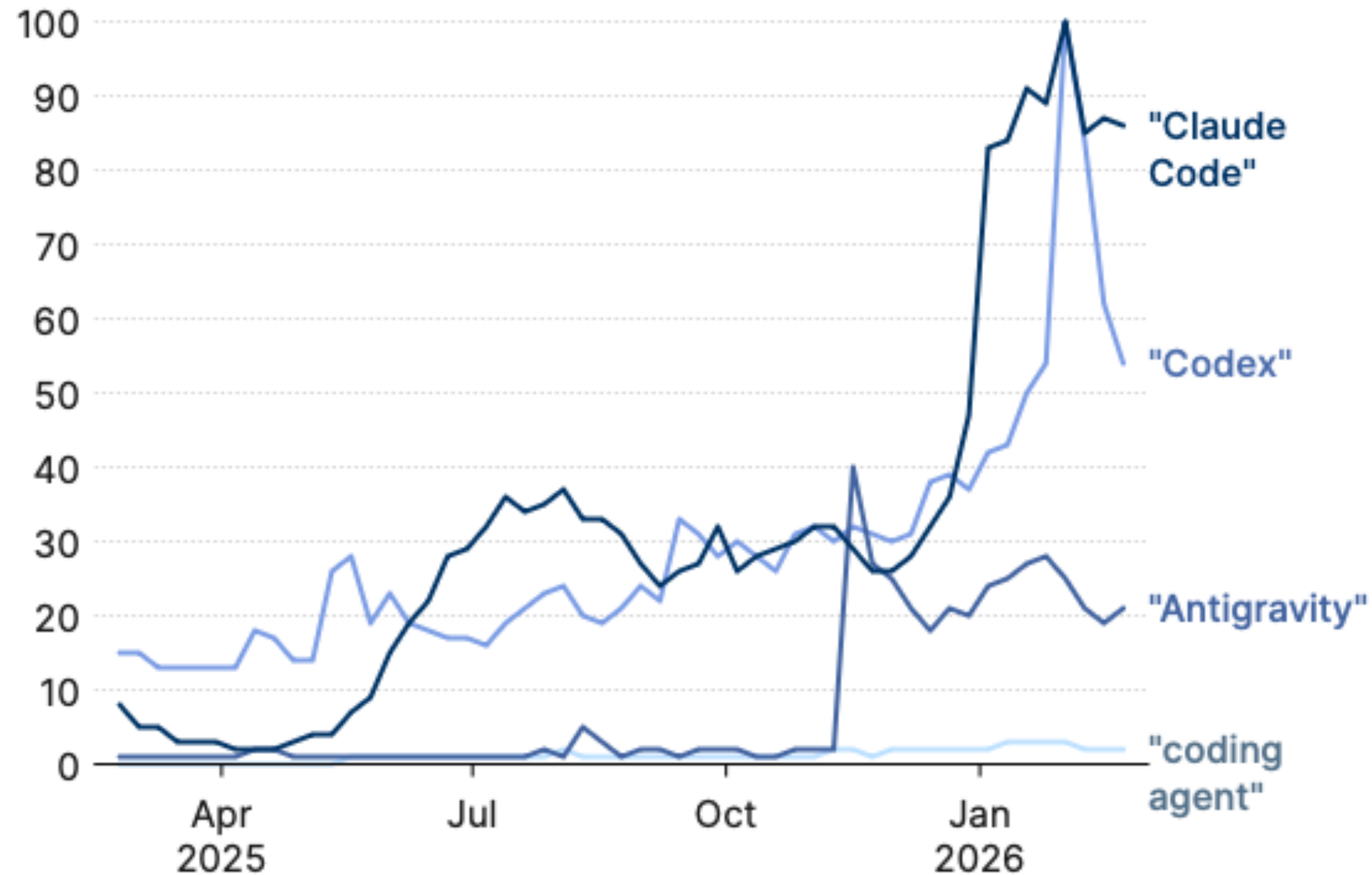
ACT 3 — The Toolkit

Your new favorite research tools

ACT 1 — THE SEDUCTION

Let's publish a paper during this presentation (?)

Yesterday's best tools are already today's relics.



Google Trends



ACT 2 — Getting up to Speed

AI - ML - LLM - GenAI - RAGs - Agents

What is « AI » ?

What society this it is



**GENERATIVE
AI**

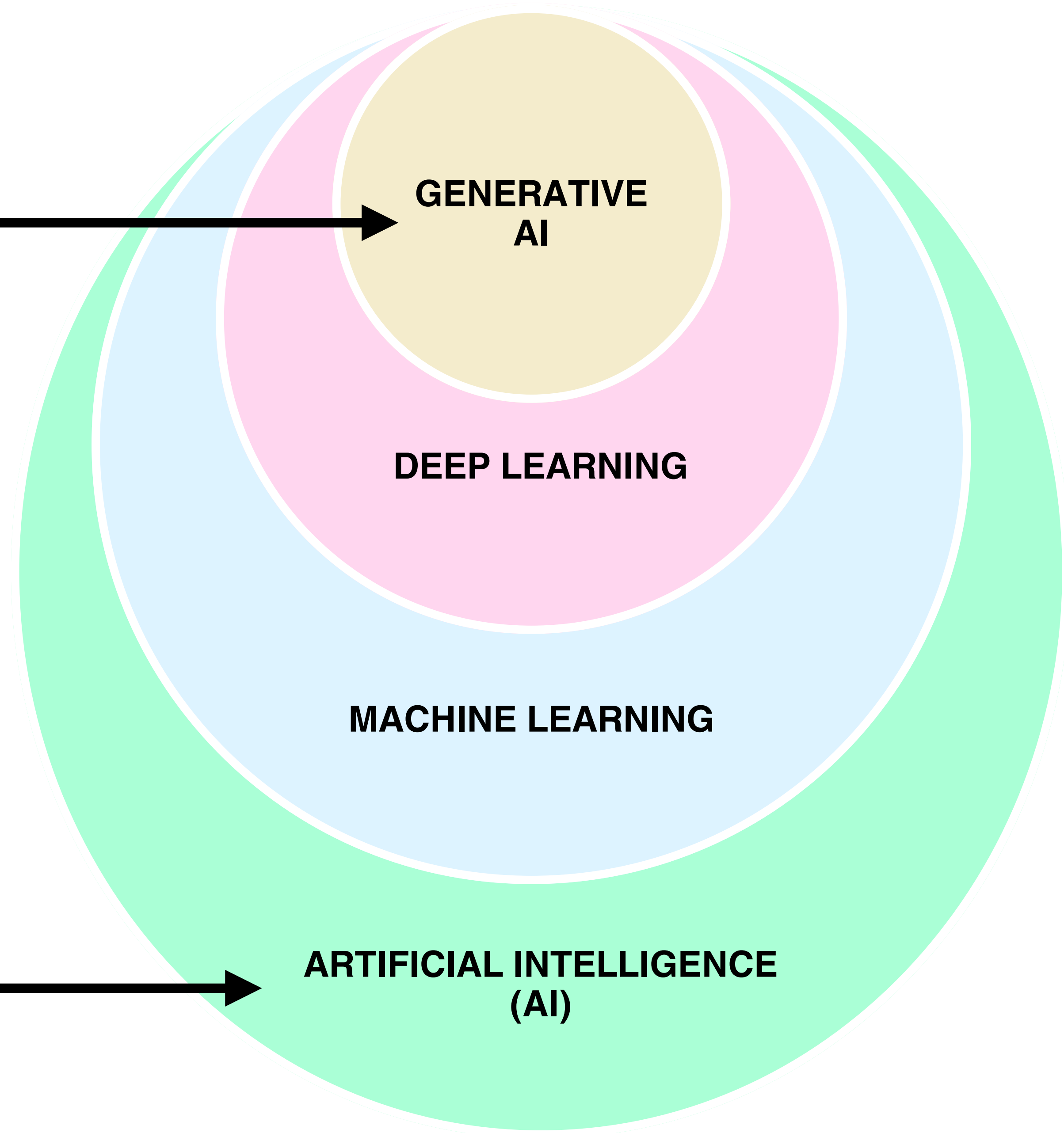
DEEP LEARNING

MACHINE LEARNING

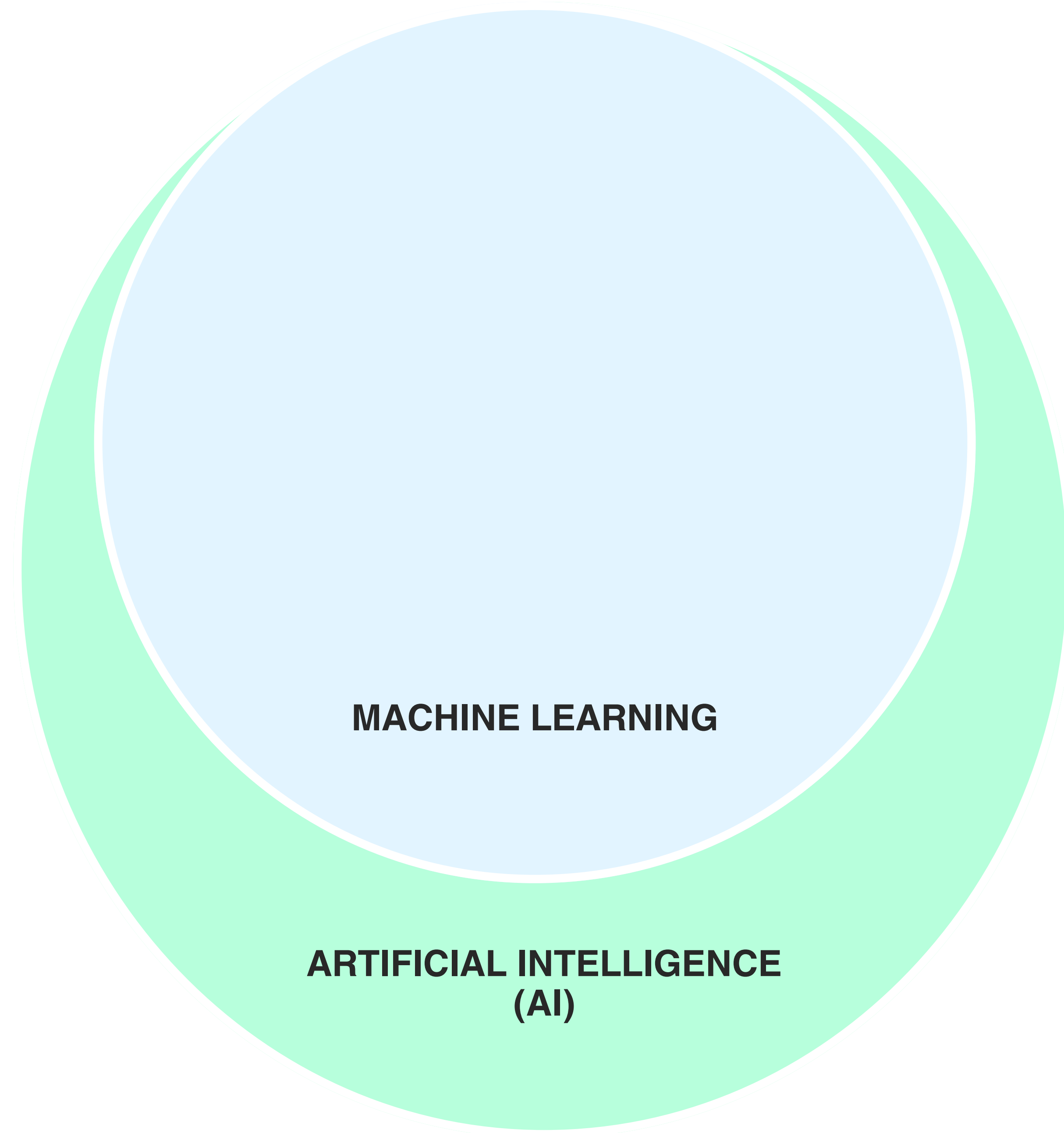
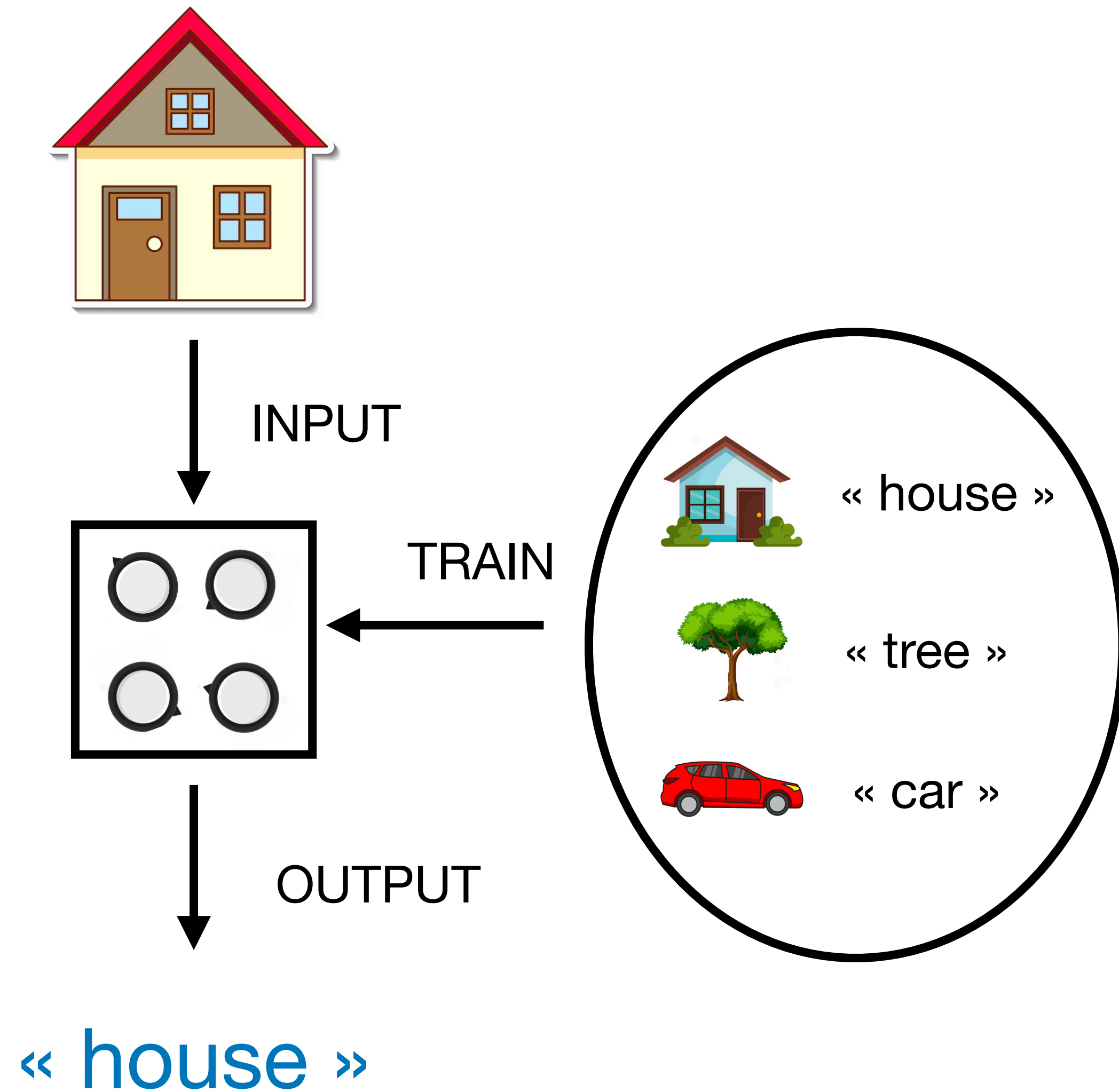
What it really is



**ARTIFICIAL INTELLIGENCE
(AI)**



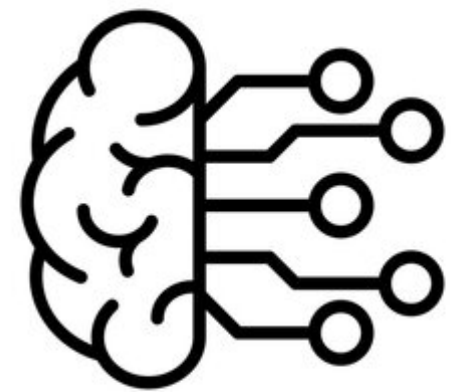
What is « ML » ?



What is an « LLM » ?

« I'm tired. »

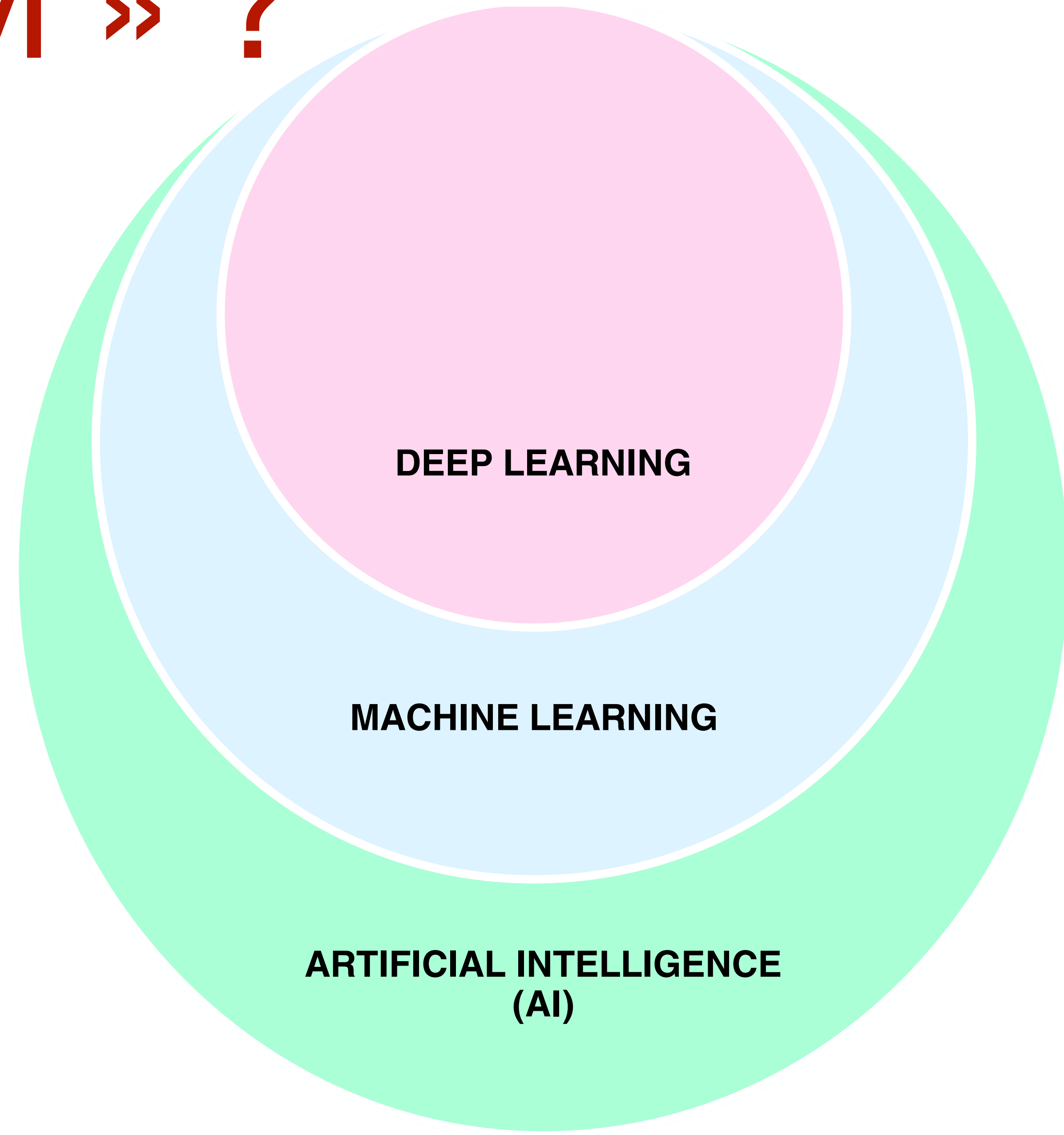
INPUT



TRAIN

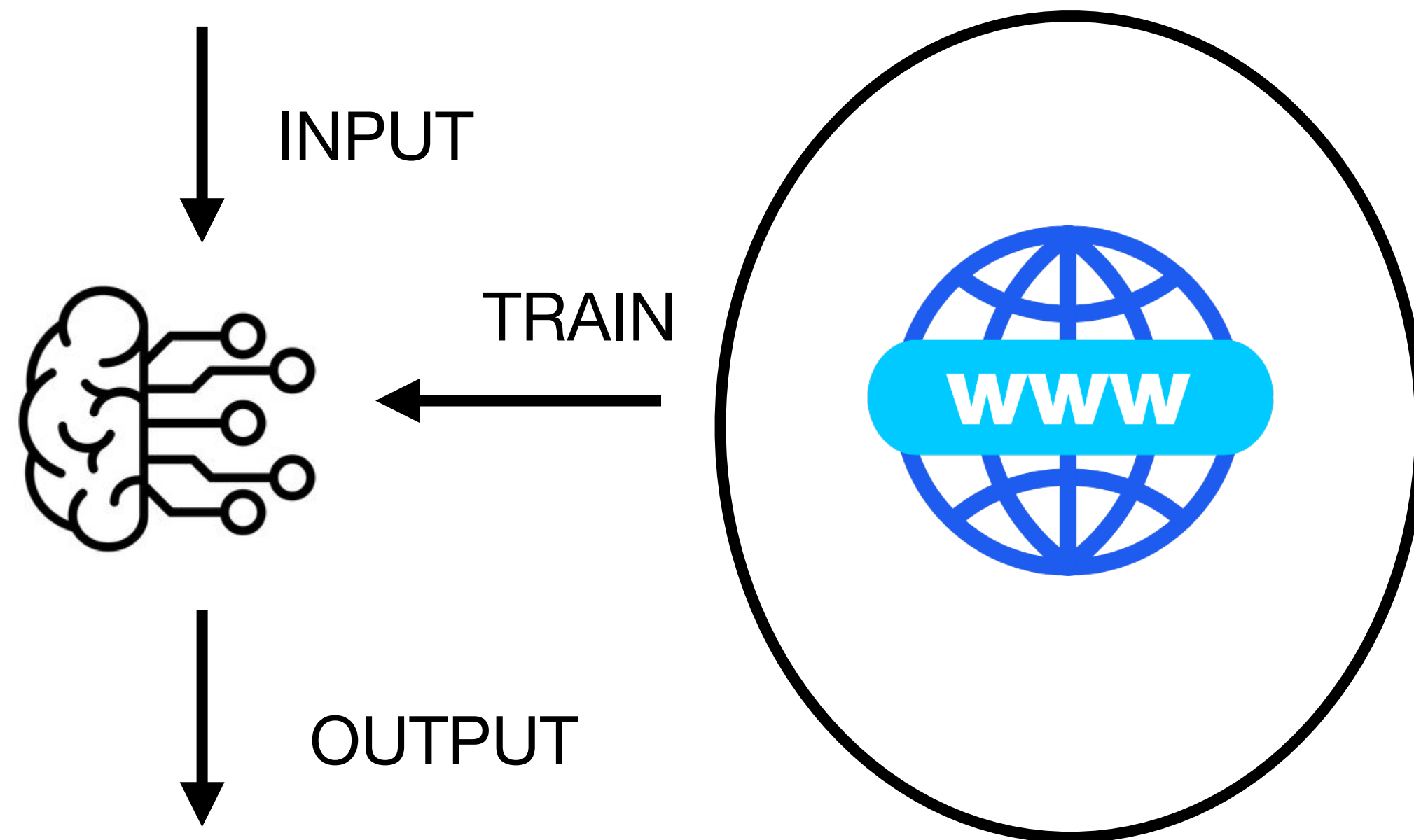
OUTPUT

« Je suis fatigué »



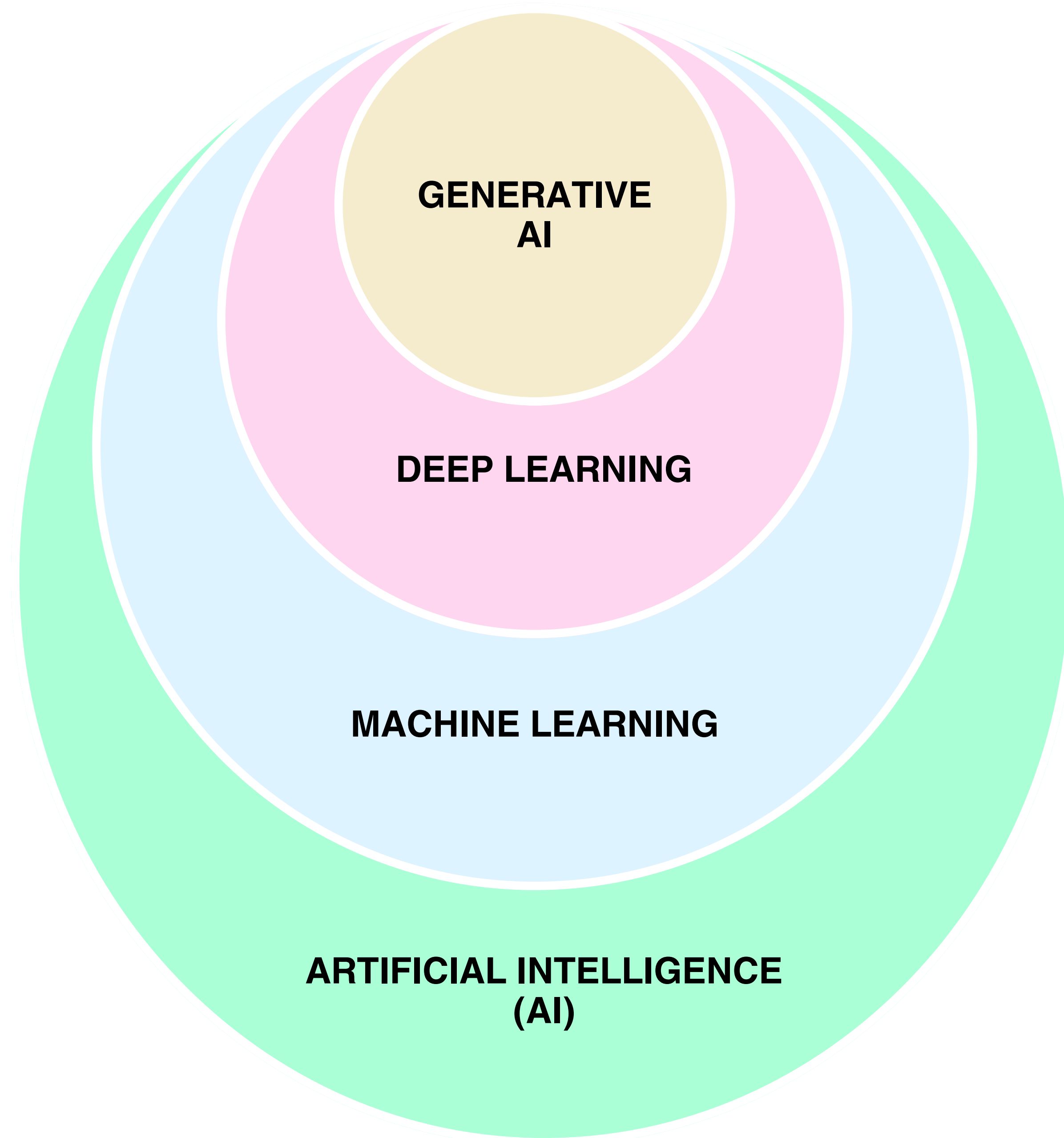
What is GenAI ?

« Paris est la
capital de ... »

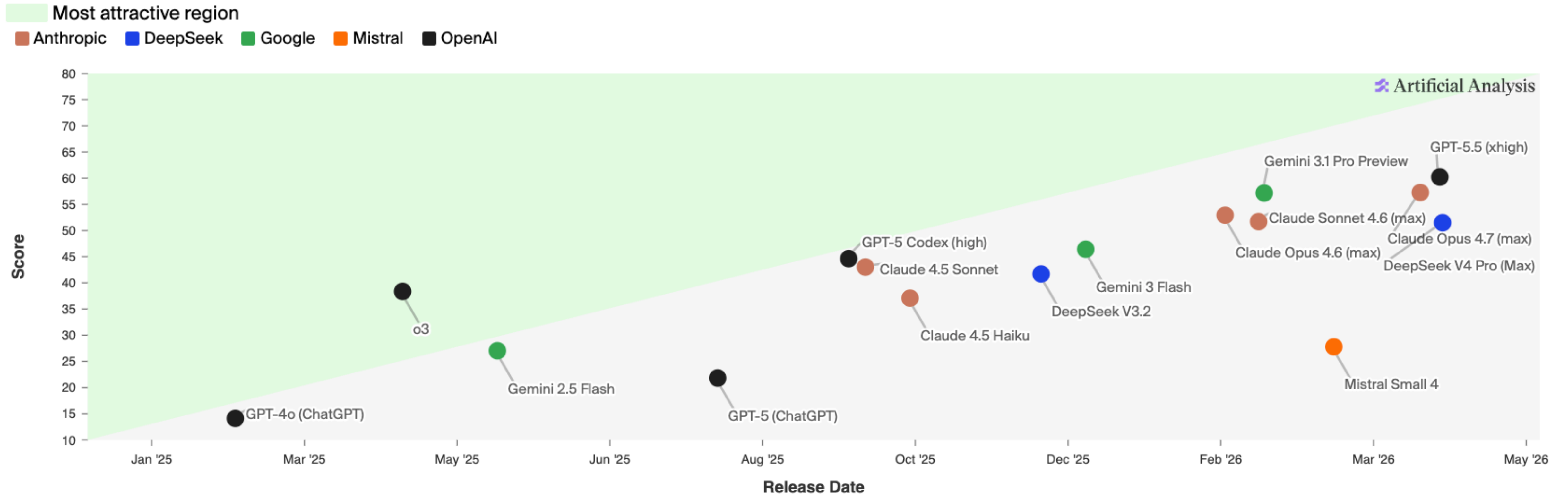


« France » : 93% chance

« mode » : 7% chance



Today's GenAI models



Decoding the AI Buzzwords



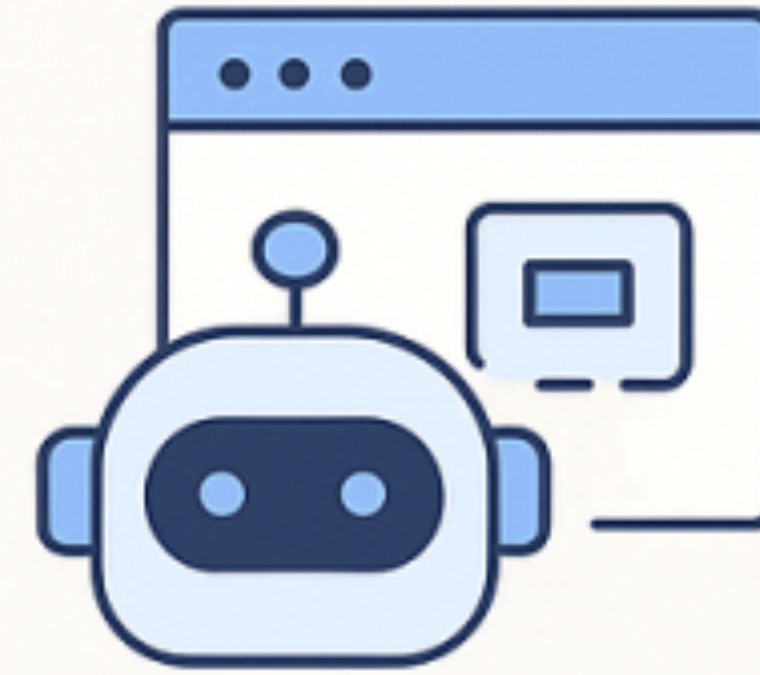
LLM

Large
Language
Model



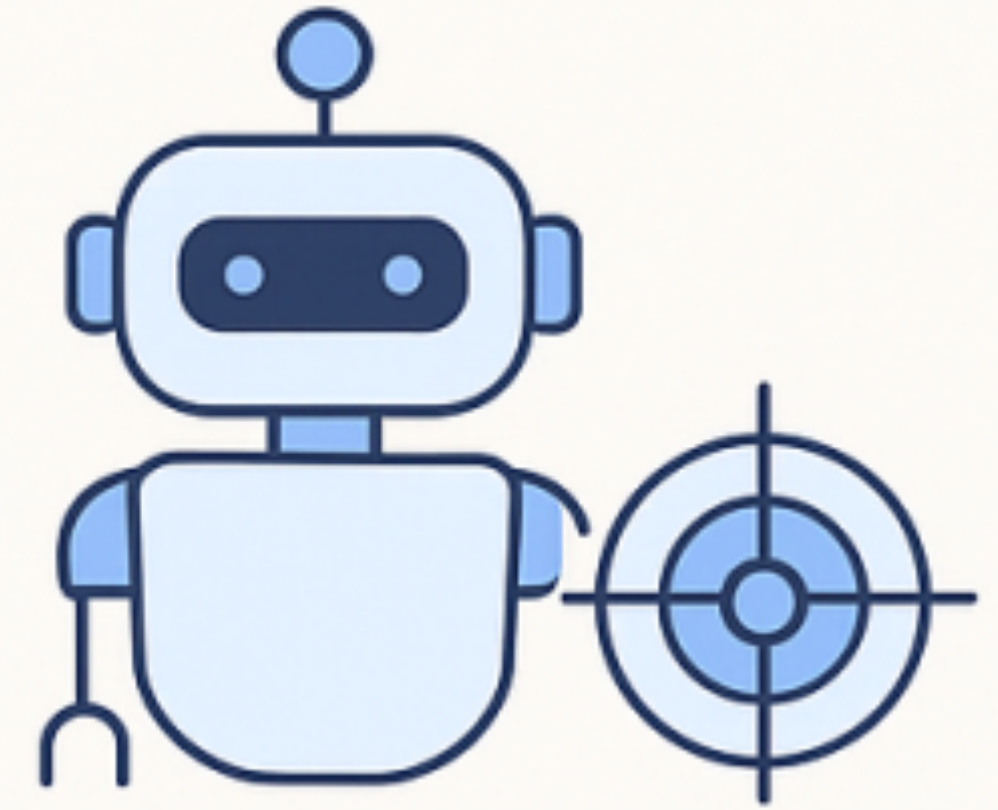
RAG

Retrieval-
Augmented
Generation



AI Agent

Executes
tasks



Agentic AI

Plans and acts
autonomously

Decoding the AI Buzzwords



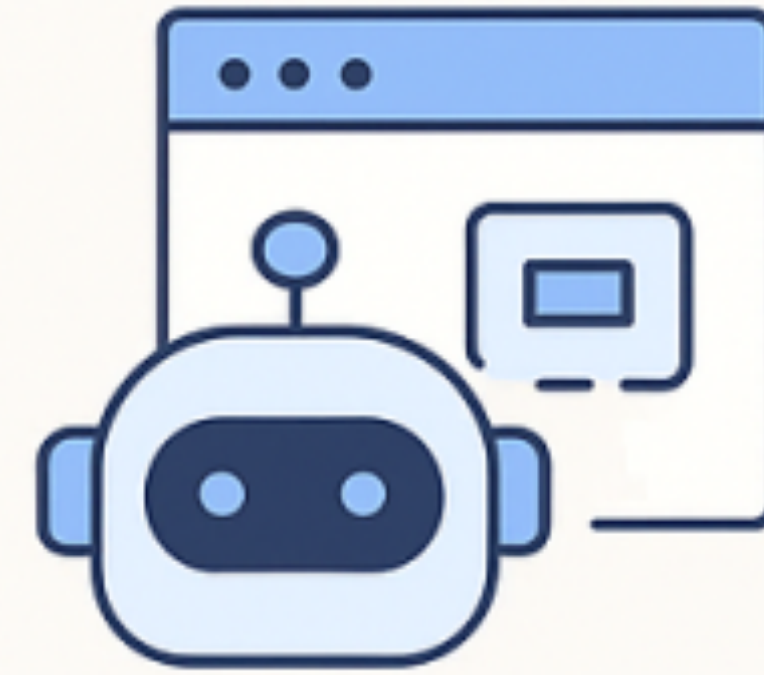
LLM

Large
Language
Model



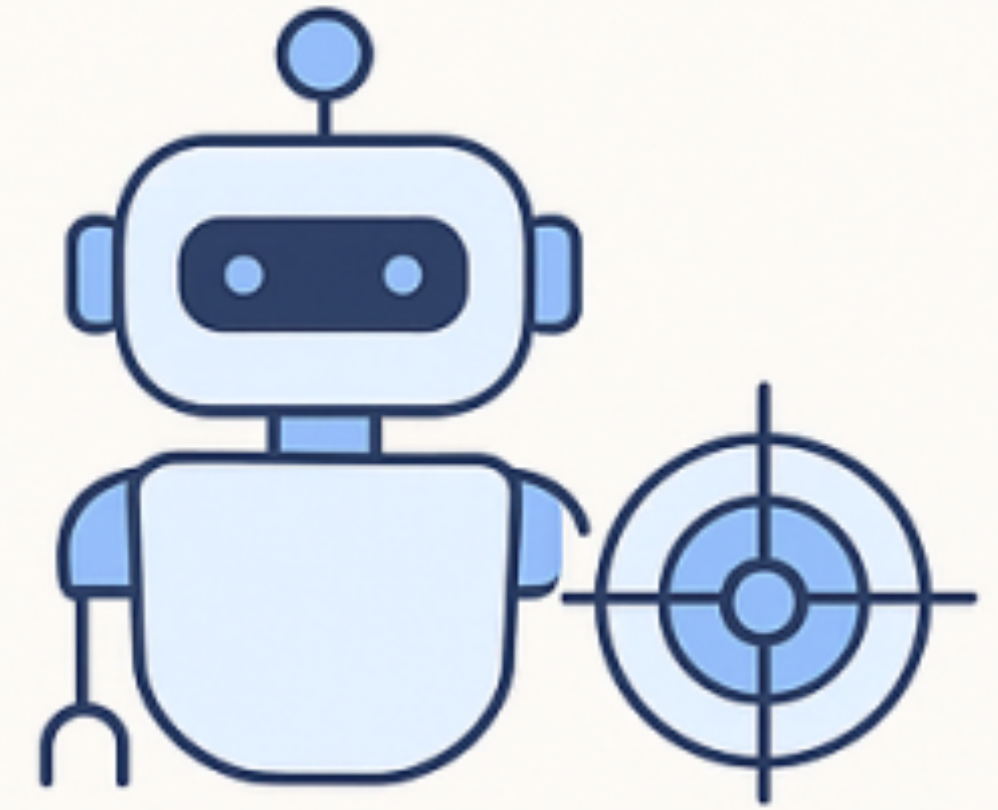
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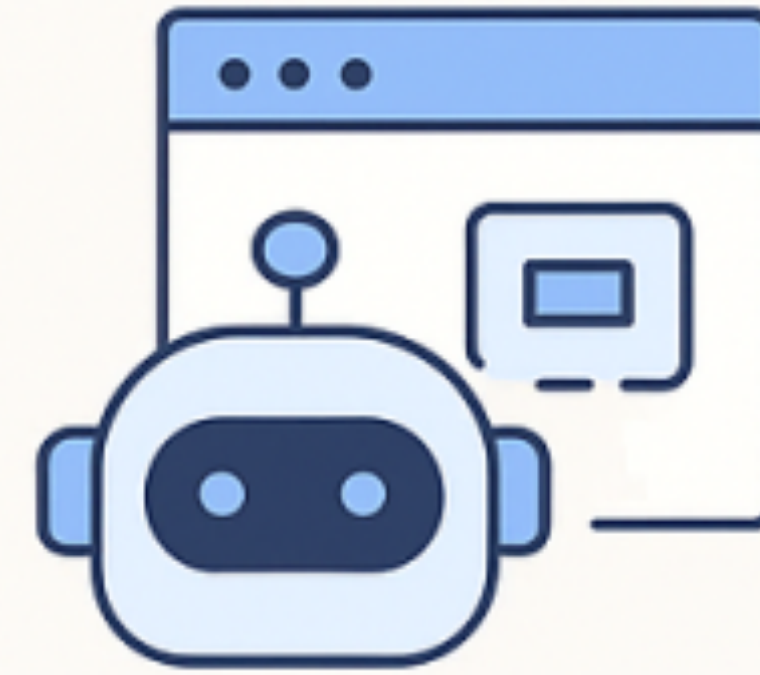
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Large
Language
Model



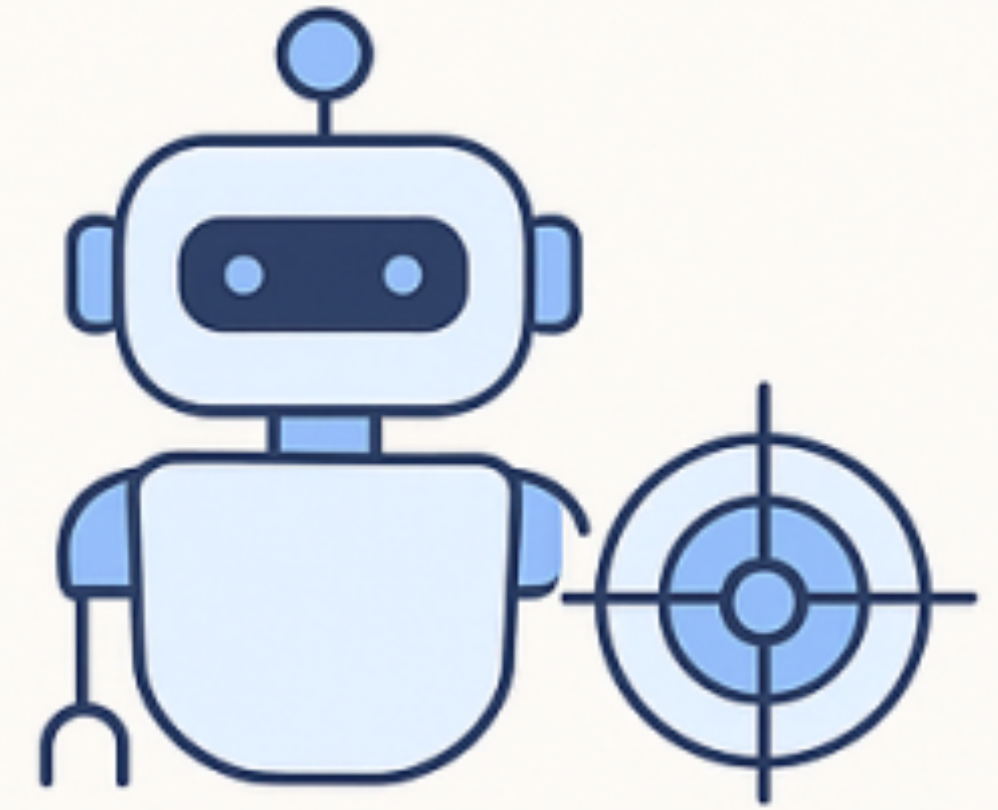
RAG

Retrieval-
Augmented
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AI Agent

Executes
tasks



Agentic AI

Plans and acts
autonomously

ACT 3 — THE TOOLKIT

Your new favorite research tools

Principles

Warning

1. AI is a copilot, not an author
It helps thinking, not replaces it
2. Garbage in → garbage out
Prompt quality = output quality
3. Always verify sources
Especially citations

explore → narrow → read → synthesize → produce

The Research Journey

Brainstorming

Literature

Hypothesis

Methodology

Data
Collection

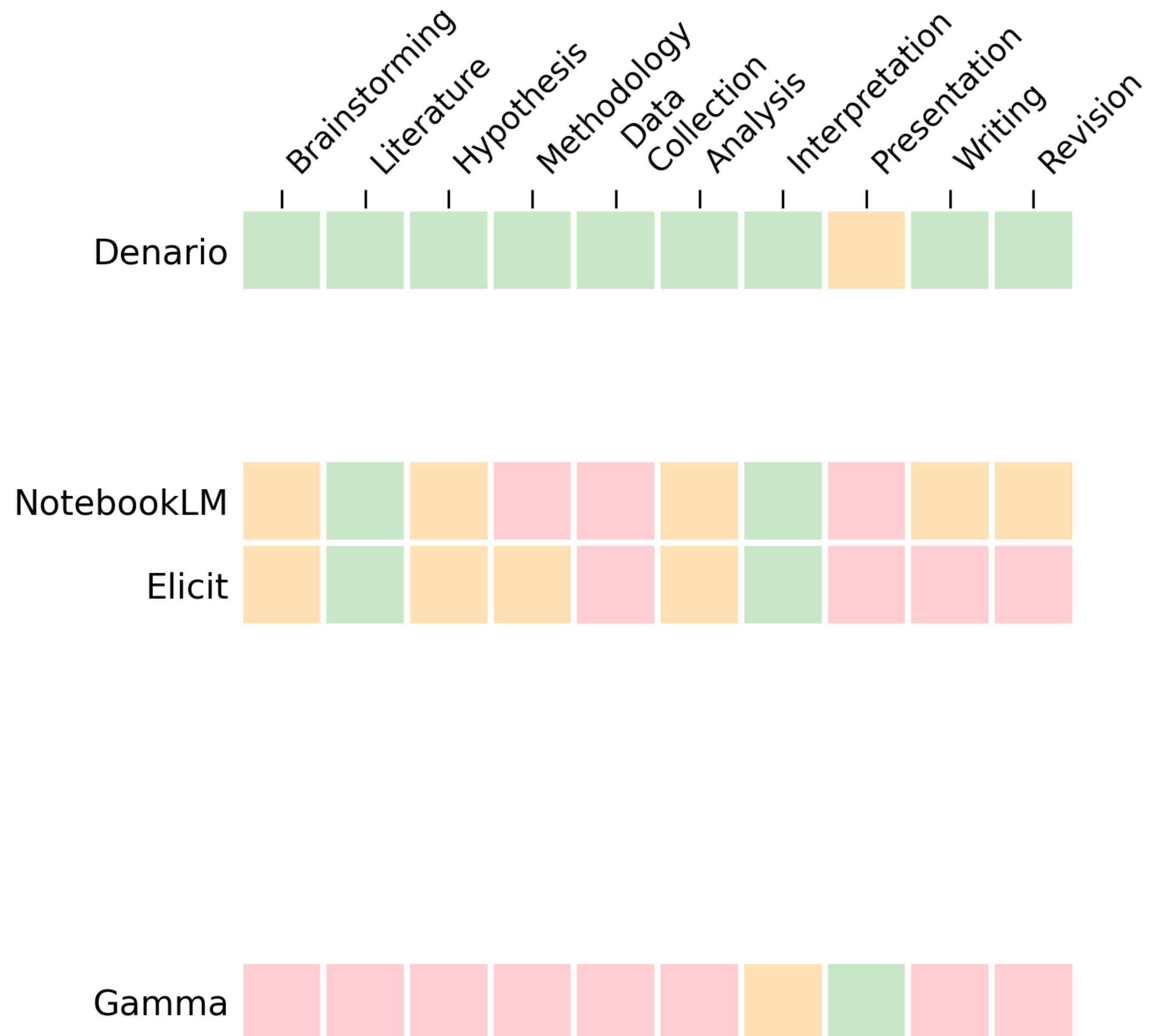
Analysis

Interpretation

Presentation

Writing

Revision



Analyze research papers at superhuman speed

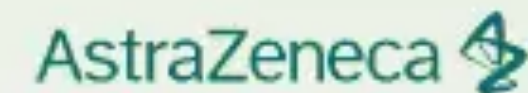
Automate time-consuming research tasks like summarizing papers, extracting data, and synthesizing your findings.

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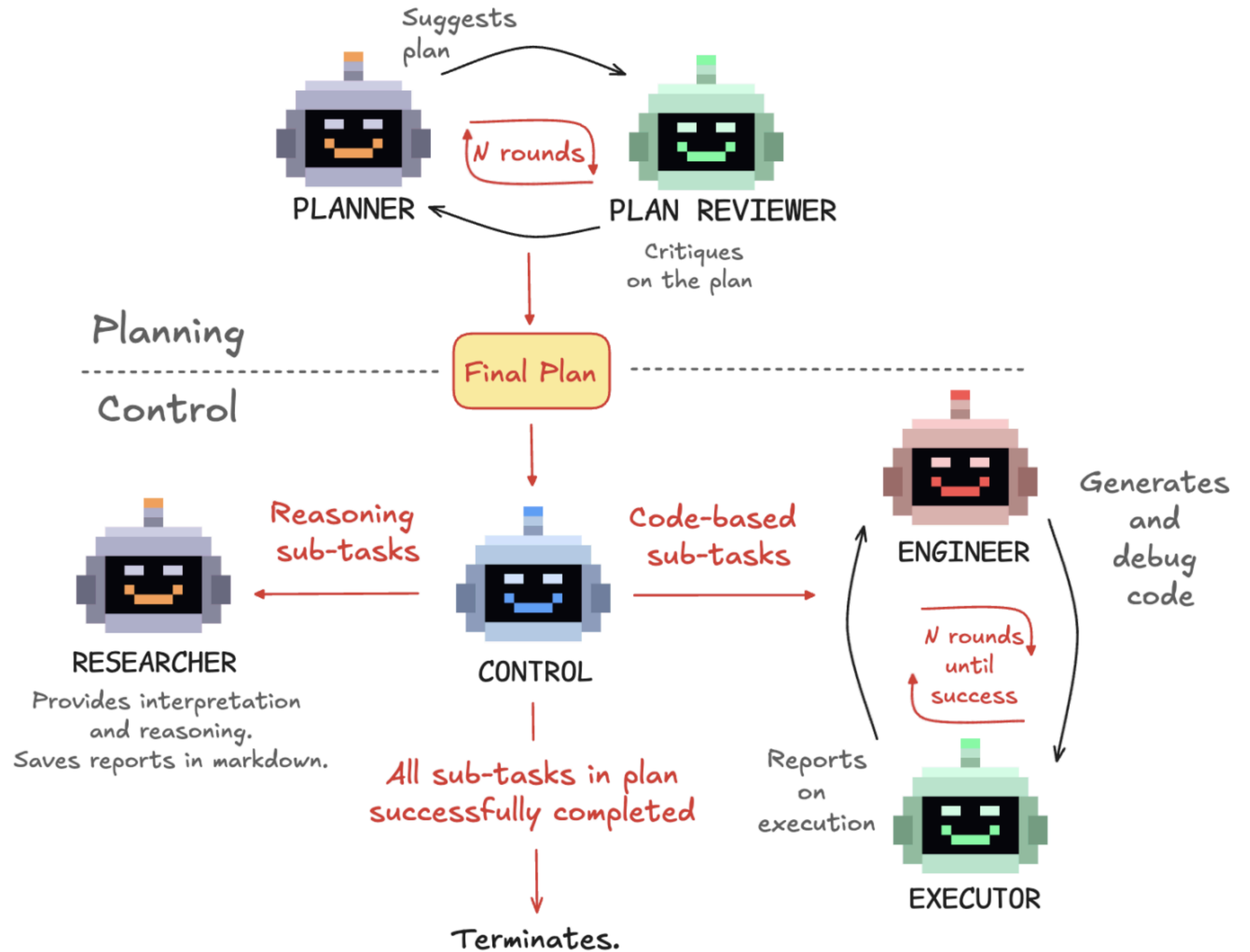
Gamma AI Review





NotebookLM

Denario



EXPLORATEUR

... Claude Code X

✓      ...

demo_2_paper

← →

DEMO_2_PAPER

.env

CLAUDE.md

funding_application.pdf

Untitled

⌚ +

 Claude Code



Make a CLAUDE.md file for instructions
Claude will read every single time.

 Prefer the Terminal experience? Switch back in Settings. X

> STRUCTURE

> CHRONOLOGIE

⌘ Esc to focus or unfocus Claude

+ ↗

| 2 lines selected

</> Edit automatically ⬆

EXPLORATEUR

... Write scientific paper f... X

DEMO_2_PAPER

.env

CLAUDE.md

funding_application.pdf

Write scientific paper from funding appli...

</> CLAUDE.md#7-8

Bonjour Claude !

En utilisant tes skills "Denario", peux-tu me créer un publication scientifique sur la thématique décrite dans "funding application" ?

Thought for 0s >

Agent: Explore project structure and funding application

IN Explore the project at /Users/cvanderkerckh/Code/demo_2_paper. I need to understand:
1. The full directory structure (ls -la recursively for key folders)
2. The content of any "funding application" file(s) - look for files named funding*, grant*, proposal*, application*, or similar

OUT

Bash List data directory

IN ls -la /Users/cvanderkerckh/Code/demo_2_paper/data 2>/dev/null | head -30

OUT (Bash completed with no output)

Bash List directories only

IN ls -la /Users/cvanderkerckh/Code/demo_2_paper/ | grep -E "^d"

OUT drwxr-xr-x 5 cvanderkerckh staff 160 Mar 25 16:05 .
drwxr-xr-x 84 cvanderkerckh staff 2688 Mar 25 16:01 ..

Bash Find all directories

IN find /Users/cvanderkerckh/Code/demo_2_paper -type d 2>/dev/null

OUT /Users/cvanderkerckh/Code/demo_2_paper

Thought for 22s >

Imagining...

Queue another message...

+ | 2 lines selected

Plan mode

> STRUCTURE

> CHRONOLOGIE

EXPLORATEUR

DEMO_2_PAPER

> paper

> results

.env

.gitignore

.python-version

analysis.py

CLAUDE.md

funding_application.pdf

main.py

pyproject.toml

run_pipeline.py

uv.lock

Write scientific paper f... X

Write scientific paper from funding appli...

</> CLAUDE.md#7-8

Bonjour Claude !

En utilisant tes skills "Denario", peux-tu me créer un publication scientifique sur la thématique décrite dans "funding application" ?

Patch mistralai in embagent

Write run_pipeline.py (Denario workflow)

Run analysis.py to generate results

Run run_pipeline.py to generate paper

Bash

Run Denario pipeline to generate paper

IN

uv run python run_pipeline.py 2>&1

OUT

/Users/cvanderkerckh/Code/demo_2_paper/.venv/lib/python3.12/site-packages/denario/utils.py:75: UserWarning: No data files paths were found in the data description. If you want to provide input data, ensure warnings.warn(
Generating NeurIPS-format paper via Denario...
Files generated: 0.1.0 Failed to run the Denario pipeline 2024-08-10 10:00:01

Bash

List generated paper and figures

IN

ls /Users/cvanderkerckh/Code/demo_2_paper/paper/ && echo "----" && ls /Users/cvanderkerckh/Code/demo_2_paper/results/figures/

OUT

input_files
paper

...

Bash

List paper output files

IN

ls /Users/cvanderkerckh/Code/demo_2_paper/paper/paper/

OUT

LLM_calls.txt
LaTeX_compilation.log
LaTeX_err.log
...

* Herding...

Queue another message...

+ / | 2 lines selected

</> Edit automatically

Take Home Message

ACT 1 — AI is already transforming research

Take Home Message

ACT 1 — AI is already transforming research

ACT 2 — Understanding remains essential

Take Home Message

ACT 1 — AI is already transforming research

ACT 2 — Understanding remains essential

ACT 3 — Tools are powerful, but not neutral

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